Part 1 Recording Script – Member 4

Dashboard: *Does drought risk reveal concerns that baseline water stress alone misses?*
Target length: **4:40** (assessors stop at 5:00). Speak at a steady pace, about 140 words a minute.

Before you press record

1. Start the app. In Terminal: `cd ~/Downloads/member4_water_drought_module`, then `source .venv/bin/activate`, then `python -m streamlit run dashboards/m4_stress_vs_drought.py`

2. Safari tab: `localhost:8501`. Zoom to 110–125% (Cmd +). Scroll to the top. Check all regions are selected and **Drought risk** is ticked.

3. Excel: open `data/source/Aqueduct40_rankings_download_Y2023M07D05.xlsx`, go to sheet **country_baseline**, select row 1 and turn on Filter (Data > Filter). Zoom to about 150%.

4. Processed CSV: open `data/processed/m4_water_stress_drought.csv` (Excel, or PyCharm). Find the **IRL** row in advance.

5. PyCharm (or Finder): project folder open, showing `dashboards/`, `scripts/m4_process.py` and `docs/m4_ai_provenance_rows.md`.

6. Recording: QuickTime > File > New Screen Recording > Options > choose your microphone. Close WhatsApp, mail and notifications (turn on Do Not Disturb).

7. Practice once with a timer. Switch windows with Cmd + Tab.

Do the Canada check yourself before recording (30 seconds), because you say it in the AI section. In Excel, filter gid_0 = CAN, indicator_name = drr, weight = Tot: the score is -9999. In the processed CSV, Canada's drr_score is blank.

0:00 – 0:30 · Question and persona

Checklist: question stated, persona named

What to show / do	What to say
Safari, top of the app. Point the mouse at the title, then at the Group question and Persona lines.	“Hi, I'm Harsh, Member 4. Our group's question is: **should data-centre expansion supporting AI be assessed against local electricity and water constraints rather than global averages?** Our persona is **an infrastructure policy adviser** who has to decide whether expansion guidelines should vary by location. My dashboard asks a narrower question: does drought risk reveal water concerns that baseline water stress alone misses?”

0:30 – 1:45 · My dashboard and interactions

Checklist: own contribution on screen

What to show / do	What to say
PyCharm/Finder for about 5 seconds: point at `dashboards/m4_stress_vs_drought.py` and `scripts/m4_process.py`.	“This is my module: the dashboard file and the processing script I built for our shared repo.”
Safari: the three big numbers under “Across all countries with both scores”.	“The app opens on the whole dataset. WRI's Aqueduct 4.0 gives 138 of 192 countries both a water-stress score and a drought-risk score. Of the 86 countries with low or low-medium water stress, under 20 per cent, **65, which is 76 per cent, still have medium or higher drought risk.** And the two scores are only weakly related, a correlation of 0.26.”
Scroll down to the **scatter chart**. Move the mouse along the dashed lines, then circle the shaded top-left zone.	“Each point is a country. The dashed lines are Aqueduct's own category boundaries at score 2. The shaded top-left zone means low water stress but elevated drought risk. Ireland, the UK, the Netherlands, France and Japan, all big data-centre markets, are in it.”
Interaction 1: in UN region, remove every region except Europe (click the small x's). Wait for the chart to update.	“If I filter to Europe, the chart, the takeaway text and the table all update.”
Interaction 2: click **Riverine flood risk**.	“I can also switch the second indicator to flood risk, which is more of a siting risk than a water supply risk.”
Empty state: click the grey (x) in the region box to clear everything. Show the blue message. Then re-add all regions and click **Drought risk** again.	“And if nothing is selected, the app shows a message instead of breaking.”

1:45 – 2:35 · Design decision

Checklist: design decision with a named alternative and why it lost

What to show / do	What to say
Safari, stay on the scatter. Point at the labelled blue dots, then scroll a little to show the table.	“The visual task here is a **relationship**: finding countries where two measures disagree. So I chose a **scatter plot**. The alternative I rejected was **two ranked bar charts side by side**, one for water stress and one for drought. I rejected it because the reader would have to find each of 138 countries in two separate lists to spot a mismatch. In a scatter, every country is one point, so the mismatch shows up as a zone you can see straight away. I also rejected a combined risk score, because averaging the two would hide exactly the disagreement my question is about. Key countries are labelled directly and there's a table underneath, so you don't need to hover to read any value. Labelled points differ in fill as well as colour, so it still works for colour-blind viewers.”

2:35 – 4:05 · Trace a number live

Checklist: a displayed number traced back to the source data

What to show / do	What to say
Safari: scroll to the table. Point at **Ireland, drought risk score 2.62**.	“Now I'll trace one number. Ireland's drought risk score is **2.62** in the table.”
Excel, sheet country_baseline. Filter **gid_0 = IRL**, then **indicator_name = drr**, then **weight = Tot**. Point at the score, cat and label cells.	“This is WRI's original download. Filtering to Ireland, drought risk, total-demand weight, the score is **2.617392**, category 2, labelled ‘Medium, 0.4 to 0.6’. I use the ‘Tot’ weight, which weights by total water demand, the same weighting as the water-stress score.”
PyCharm: open `scripts/m4_process.py` for 5 seconds (top comment block). Then the **processed CSV**, IRL row, **drr_score** column.	“My script keeps that row, turns the minus-9999 ‘no data’ code into a blank, not a zero, and reshapes the data to one row per country. In the processed file, Ireland's drr_score is the same: 2.617392.”
Safari: scroll down, open **Trace a number to the source file**. Ireland is already selected. Point at the source rows, then the processed row.	“The app also shows both rows side by side. On screen it's rounded to **2.62**. That's above 2, the Medium boundary, and Ireland's water-stress score of 0.36 is below 2, so Ireland gets flagged as low stress but elevated drought risk.”

4:05 – 4:40 · AI use and the check

Checklist: AI use disclosed, with the check applied

What to show / do	What to say
PyCharm: open `docs/m4_ai_provenance_rows.md`.	“I used Claude, an AI assistant, to inspect the workbook and draft the processing script and the Streamlit app. It pointed out that the file has no seasonal-variability indicator. I confirmed that myself by listing the indicators, and then switched to drought risk. For the script, my check was that the output must have exactly one row per country, which is 192. **The first version failed that check**: it produced over a million rows because of a pivot setting. It was fixed and re-run, and I also checked by hand that Canada's missing drought score stayed blank instead of becoming zero. Thanks.”

Stop recording at about 4:40.

Final checklist before uploading

[ ] Under 5:00
[ ] Voice clear, screen readable
[ ] Group question said
[ ] Persona named
[ ] My own files and dashboard shown
[ ] Scatter vs two bar charts, with the reason
[ ] Ireland 2.62 traced to the Excel file
[ ] AI use explained, plus the check that failed

Upload to Canvas as a file or an unlisted YouTube link. Only say the AI lines you can stand behind.